

Aaron Teng

(408)-709-6460 — [Personal Website](#) — aaronteng@berkeley.edu — [LinkedIn](#) — github.com/ar1s1502

Education

Computer Science, B.A. U.C. Berkeley
GPA: 3.86; expected graduation: Spring 2027

August 2023 – Present

Projects

Cappella Catalogue — <https://cappellacatalogue.com>

[Github](#)

- Online sheet music store for a cappella arrangements. Full-stack web app built in Django, HTML/Bootstrap/JS, and PostgreSQL, with a live Stripe integration for payment, deployed on an AWS EC2 Ubuntu instance using Nginx, Gunicorn, and Docker Compose.

RuSH

[Github](#)

- POSIX-compliant shell program in Rust that can run in a REPL or as a shell-script interpreter, featuring a parsing engine and execution loop that identifies syntax errors and prompts the user for continuation.
- Features a state-machine-based lexer that understands and enforces shell-script syntax and tokenizes via regex, an LL(1) recursive-descent parser, and an AST-walking execution engine.
- Supports Unix shell operators, variables, command substitutions, subshells, arbitrary chaining of heredocs, pipelines, redirection, etc. with performance comparable to standard shells like Bash or Zsh.

SecureStore

[Github](#)

- Used cryptography functions such as block ciphers, RSA key-sharing, and hashing to implement a secure file-sharing system over a simulated insecure, public database. Written in Go, with BDD test suite in Ginkgo/Gomega.
- Implementation scored in top 1.3% of class, for Berkeley's cybersecurity course (CS161).

Experience

Software Developer Intern IBM

June 2026 – September 2026

- Architected and implemented an automated backup-restore framework for IBM CloudPak For Business Automation (CP4BA), a microservice-based SaaS suite deployed on Kubernetes/OpenShift clusters.
- Engineered dynamic deployment detection to automatically target and back up mandatory Kubernetes resources and database state (PostgreSQL, IBM Db2), then fully wipe and restore the CP4BA namespace from backup.
- Reduced end-to-end backup-restore time from ~5 hours (manual) to 1 hour by integrating the automation into a Jenkins CI/CD pipeline, and further optimized backup runtime by 50% with multithreading via Goroutines.

Research Intern Berkeley SKY Computing Lab – ADRS

October 2025 – December 2025

- Built Python frameworks using the GEPA API that leverage AI models to design, optimize, and evaluate complex algorithms through a continuous design-test-feedback loop.

Optician Simply Vision Optometry

October 2023 – August 2025

- Conducting pre-tests (refraction, stereopsis, etc.) during patient intake, fitting and styling frames and lenses, billing insurance, filing patient records.
- 12–15 hours per week throughout the school year and summer.

Piano Teacher Pacific Piano School

July 2022 – July 2023

- Gave private lessons to students enrolled in summer and winter camps to teach music theory and refine their piano technique.

Skills

Technical: Python, Rust, Kubernetes, Docker, Django, Java, C, AWS, HTML/CSS/Javascript/Typescript, RISC Assembly, Go/Golang, Full-Stack Web Development, Kubernetes, Openshift, RedHat Enterprise Linux, Systems Programming

Music: Music composition (Chamber ensemble, choir), Piano accompaniment

Coursework

- | | |
|---|---|
| • Data Structures and Algorithms | • Computer Security |
| • Great Ideas of Computer Architecture | • Intro to Software Engineering |
| • Discrete Mathematics and Probability Theory | • Intro to Linux, SysAdmin |
| • Linear Algebra | • Operating Systems and Systems Programming |
| • Efficient Algorithms and Intractable Problems | • Internet Architecture and Protocols |